6th class

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Flag Register

micro-controller এ total 16 টা flag Register থাকে। প্রত্যেকটা flag 1 bit করে

16 Bit → 6 টা status flag
→ 3 টা control flag
→ 7 টা unused

Status flag:

Status Flag গুলো mainly SUB and Add এর জন্য কাজ করে।

The status flags reflect the result of an instruction executed by the processor

কোনো Arithmetic operation (→ Add and SUB) এর ~~nature~~ Result এর nature বুঝা যায় status flags দিয়ে।

i) Zero flag (ZF): (Add | Sub)

ZF = 0, যদি result of Arithmetic is non zero

ZF = 1, যদি result of Arithmetic is zero

Ex:

$$\begin{array}{r} 1A\ H \\ +\ 2C\ H \\ \hline 46\ H \end{array}$$

1A H → Hexa decimal

46 H → Result ≠ 0 so, ZF = 0

ii) Sign Flag (SF)

SF = 0, মানে Result positive
SF = 1, মানে Result Negative.

এক্ষেত্রে আমরা Result এর **16th Bit** এর Binary বসাতে পারি।

N.B Result এ যদি 1 carry Bit ও থাকে তাও 16th number bit এর check দিবো।

Sub and Add হবার ক্ষেত্রে same rules.

Ex:

$$\begin{array}{r} 8000\ h \\ +\ 0001\ h \\ \hline 8001\ h \end{array}$$

1000 0000 0000 0001

Negative so, SF = 1

Ex:

$$\begin{array}{r} 8000\ h \\ -\ 0001\ h \\ \hline 7FFF\ h \end{array}$$

0111 1111 1111 1111

positive so, SF = 0

iii) carry flag (CF)

CF = 0, মানে result 16 Bit
CF = 1, " result 16 Bit এর থেকে বড়

N.B Sub এর জন্য shortcut, যদি ~~Large~~

Large - small, cf = 0
Small - Large, cf = 1 } Always.

Large - small, CF = 0

Ex:

$$
\begin{array}{r}
CFB1\ h \\
+\ ABCD\ h \\
\hline
1\,7B7E\ h
\end{array}
$$

0001 0111 1011 0111 1110

17th Bit, so, CF = 1

Remember যেহেতু microprocessor হ 16 bits এর তাই যদি carry আসেই 1-ই আসবে

Ex:

$$
\begin{array}{r}
E412\ H \\
-\ 7B7E\ H \\
\hline
6894\ H
\end{array}
$$

0000 0110 1000 1001 0100

17th Bit so, CF = 0

iv) Parity flag (PF)

Result এর LSB থেকে Eight (৮) টা Bit check নিয়ে এর 1's count করবো

PF = 0, if 1's count is ~~even~~ odd

PF = 1, if 1's count is ~~odd~~ even

Ex:

$$
\begin{array}{r}
CFB1\ h \\
+\ ABCD\ h \\
\hline
1\,7B7E\ h
\end{array}
$$

0111 1110

1's count even → 6

so, PF = 1

Ex:

$$
\begin{array}{r}
E412\ h \\
-\ 7B7E\ h \\
\hline
6894\ h
\end{array}
$$

1001 0100

1's count odd → 3

so, PF = 0

v) Auxiliary flag (AF):

Hexa তে LSB হতে যে যোগ / বিয়োগ করবো।

~~যদি result~~

AF = 0, if result 4 Bit এই আসে (carry নাই)

AF = 1, " " 4 Bit এর - " (carry আছে)

~~Short~~ Shortcut for SUB

→ $Large_{LSB} - Small_{LSB}$, AF = 0

→ $Small_{LSB} - Large_{LSB}$, AF = 1

Ex:

```
  CFB1 h
+ ABCD h
```

1 + D = E h

1110

→ carry নাই so, AF = 0

Ex:

```
  E412 h
- 7B7E h
```

~~2-E = 10 H~~

2 - E = FFF4 H

1111 1111 1111 0100

carry আছে, so, AF = 1

অন্য way তেও বুঝা যাইতে পারে

Small - Large, so, AF = 1

= 2 - E

N.B 2 + E = 10 h → 10 h = 16 Decimal

10000

→ carry so, AF = 1

OF = 0, means register এর value valid (কারণ কোন Bit overflow হয় নাই) অর্থাৎ register 16 Bits result এ 16 bits

vi) Over flow Flag : (OF) :

২টা sign এর number add/sub করলে যেই nature এ আমার কথা যদি সেই nature এ না আসে তাহলে OF = 1, Otherwise OF = 1

Ex:

1000 (−ve)

$$\begin{array}{r} 8000h \\ (+)\ 8000h \\ \hline 10000 \end{array}$$

1000 (−ve)

carry বাদ

0000 (+ve)

(−ve) + (−ve) = (−ve) আসার কথা ছিলো কিন্তু (+ve) আসছে that means nature change so, OF = 1

Ex:

1010 (−ve)

$$\begin{array}{r} ABCDh \\ (+)\ CFB1h \\ \hline 17B7Eh \end{array}$$

1011 (−ve)

0111 (+ve)

(−) + (−) = (−ve) কিন্তু result (+ve) so, OF = 1

Ex:

$$\begin{array}{r} 5 \\ (-)\ 4 \\ \hline 1 \end{array}$$

5 → 0101 (+ve)

4 → 0100 (+ve)

0001 (+ve)

২টা (+ve) কিন্তু বড় (+ve) থেকে ছোট (+ve) sub করলে result (+ve) ই আসার কথা আসছেও so, OF = 0

Ex:

$$\begin{array}{r} 4 \\ -5 \\ \hline FFFF \end{array}$$

4 → 0100 (+ve)

5 → 0101 (+ve)

1111 (−ve)

ছোট (+ve) থেকে (+ve) sub করে result (−ve) হয়েছে so result (−ve) হওয়ার কথা, Nature আসার কথা ঠিক আছে so, OF = 0

Shortcut:

Add

if, (+ve) + (+ve) = (−ve) and (−ve) + (−ve) = (+ve)

OF = 1, Others সব case এ OF = 0

Sub

if, (−ve) − (−ve) = (−ve) and (−ve) − (+ve) = (+ve)

OF = 1, Others সব case এ OF = 0

inshort of status flag:

Flag	Add	SUB
zero flag (ZF)	ZF = 0, if Result ≠ 0 ZF = 1, if " = 0	Same
Sign Flag (SF)	SF = 0, if Result (+ve) SF = 1, if Result (-ve)	same
carry Flag (CF)	CF = 0, if 17th Bit = 0 CF = 1, if 17th Bit = 1	Large - Small, CF = 0 Small - Large, CF = 1
parity Flag (PF)	PF = 0, if count of 1 in rightmost 8 bit odd PF = 1, if count even	same
Auxiliary Flag (AF)	AF = 1, if LSB Hexa produce carry → sumadation rightmost hexa is > F AF = 0, otherwise	$Large_{LSB} - Small_{LSB}$, AF = 0 $Small_{LSB} - Large_{LSB}$, CF = 1
overflow Flag (OF)	(+ve) + (+ve) = (-ve) (-ve) + (-ve) = (+ve) OF = 1 others: OF = 0	(+ve) - (-ve) = (-ve) (-ve) - (+ve) = (+ve) OF = 1 others: OF = 0

fall 25 Q2(a)

N.B যেহেতু sequentially বলছে so, (i) এর Bx এর value দিয়ে (ii) Answer করতে হবে।

Sub/Add Bx Ax মানে Bx = Bx ± Ax

Sub/Add Ax Bx " Ax = Ax ± Bx

আমরা যেটা আগে বলবো সেটাতে Result overwrite হবে → সেক্ষেত্রে old value মুছে যাবে।

(i) Add Bx Ax

Bx: CFB1 h
Ax: ABCD h
17B7E h

0001 0111 1011 0111 1110

ZF = 0
SF = 0
CF = 1
PF = 1
AF = 0
OF = 1

(ii) এখানে Bx: CFB1 h নয় Bx: 7B7E (not Bx: 17B7E)

Sub Cx Bx

Cx: E912 h
Bx: 7B7E h
Cx: 6894 h

0110 1000 1001 0100

ZF = 0
SF = 0
CF = 0
PF = 0
AF = 1
OF = 1

class - 7th

Rakib Hossain

Introduction to 8086 micro-processor

এই chapter এ ~~µp লিখবো~~ shortcut এ micro-processor কে µp লিখবো।

i) 8086 µp এ total 40 টি pin আছে
- → 16 bit µp
- → Address Bus 20 bit
- → Data Bus 16 bit
- → ALU / General purpose register 16 bit → AX, BX, CX... registers

ii) Mainly Alu দিয়ে computer কত Bit decide করা হয় (Decide the number of bit of CPU)

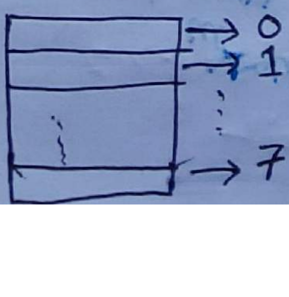
iii) একটি CPU কত Bit decide করে যে সর্বোচ্চ কত size এর RAM লাগানো যাবে।

iv) For N bit address Bus, total 2^N Location can be addressed.

Ex: suppose $n = 20$,
2^{20} or 1m addressable Locations for 8086 µp.

N.B

3 bit address bus



3 bit address Bus মানে 8 টি Location কে address বা point করতে পারবে এক সাথে।

v) যদি প্রত্যেকটা Location এ ~~Abit~~ n bit Data রাখা যায় তাহলে Total addressable size হবে

~~Total addressable size~~

Address Bus bit (AB)

CPU capacity = (# address location) * (width of Data bus)

Data Bus bit (DB)

~~$= 2^n \cdot D$~~

so, if n bit AB
m bit DB

$\boxed{\text{AB/DB always bit}}$

$$\text{CPU capacity} = 2^{AB} \cdot DB = 2^n \cdot m \text{ (bit)}$$

$$= \frac{2^n \cdot m}{8} \text{ Byte}$$

$$= \frac{2^n \cdot m}{8 \times 2^{10}} \text{ KB}$$

$$= \frac{2^n \cdot m}{8 \times 2^{20}} \text{ MB}$$

$$= \frac{2^n \cdot m}{8 \times 2^{30}} \text{ GB}$$

Ex:
suppose 20 bit AB, and 16 bit DB

$$\text{CPU capacity} = \frac{2^{AB} \cdot DB}{8 \times 2^{20}} \text{ mB}$$

$$= \frac{2^{20} \times 16}{8 \times 2^{20}}$$

$$= 2MB$$

so, 2MB RAM handle করতে পারবে max^m.

N.B
- 8086 MP এর সাথে by default 1MB RAM connect করে রাখা হয়।

RAM এর AB/DB CPU এর AB/DB এর থেকে বড় হতে পারবে না।

Fall 25 Q3 (b)

① SRAM:

DB: 16 bit

~~AB~~ Addressable Location: 64k

Capacity: n?

DRAM:

DB: 8 bit

capacity: 2n?

~~AD~~ AB: ?

Solⁿ

~~SRT~~ SRAM: $2^{AB}.DB$

Capacity: 64k × 16 bit

$$= \frac{64\times 2^{10}\times 16}{8\times 2^{20}}\ MB$$

$$= 0.125\ MB$$

DRAM:

capacity: 2n

$$= 2\times 0.125$$

$$= 0.25\ MB$$

So, $(2^{AB}\cdot DB)^{bit} = 0.25\ MB$

$$\Rightarrow \frac{2^{AB}.8}{8\times 2^{20}}\ MB = 0.25\ mb$$

$$\Rightarrow 2^{AB} = (0.25 \times 2^{20})$$

$$\Rightarrow 2^{AB} = 262144$$

$\therefore AB = 18$ bit → $\log_2(262144)$

Summer 25 Q3 (b)

DRAM:

AB = 2n

DB = 2m

capacity = 64 MB

Soln

$(2^{AB} \cdot DB)$ bit = 64 MB

$\Rightarrow \frac{2^{AB} \cdot DB}{8 \times 2^{20}}$ MB = 64 MB

$\Rightarrow \frac{2^{2n} \cdot 2m}{8 \times 2^{20}} = 64$

$\Rightarrow 2^{2n} \cdot 2m = 64 \times 8 \times 2^{20}$ —— (i)

SRAM:

AB = n

DB = m

capacity = 8 KB

$(2^{AB} \cdot DB)$ bit = 8 KB

$\Rightarrow \frac{2^{AB} \cdot DB}{8 \times 2^{10}}$ ~~MB~~ KB = 8 KB

$\Rightarrow 2^{n} \cdot m = 8 \times 8 \times 2^{10}$ —— (ii)

$$\text{(i)} \div \text{(ii)} \Rightarrow \frac{2^{2n} \cdot 2m}{2^{n} \cdot m} = \frac{64 \times 8 \times 2^{20}}{8 \times 8 \times 2^{10}}$$

$$\Rightarrow 2^{2n-n} \cdot 2 = 8 \times 2^{10}$$

$$\Rightarrow 2^{n} = \frac{8 \times 2^{10}}{2}$$

$$\Rightarrow 2^{n} = 4 \times 2^{10}$$

$$\Rightarrow n = \log_2(4 \times 2^{10})$$

$$\Rightarrow n = 12$$

$\therefore n = 12$ bit $\Longrightarrow$ (ii) $\Rightarrow 2^{n} \cdot m = 8 \times 8 \times 2^{10}$

$\Rightarrow 2^{12} \cdot m = 8 \times 8 \times 2^{10}$

$\Rightarrow m = 16$ bit

SRAM

AB = 12 bit

DB = 16 bit

DRAM

AB = 2×12 = 24 bit

DB = (2×16) = 32 bit